

LESSON 1.4

EVALUATING NUMERIC EXPRESSIONS – PART 1



LESSON INTRODUCTION

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

LEARNING TARGETS

- I can evaluate numeric expressions with up to three or four operations.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

WORKING TOWARD

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.

ESSENTIAL QUESTIONS

- How can you use order of operations to evaluate numerical expressions?
- What order did you perform the operations in?

LESSON NARRATIVE

In this lesson, students review the order of operations and apply it to solve numeric expressions. Students initially sequence given steps, then they identify and correct errors in sample students' work, and finally they independently evaluate numerical expressions. Throughout the lesson, students are asked to show their work and justify their reasoning.

COMPONENT SUMMARY

1.4.1 WARM-UP (~5 MIN.)

Notice and wonder about Castillo de San Marco

1.4.3 GUIDED PRACTICE (10 MIN.)

Reorder and explain correct order of operations

1.4.5 ACTIVITY (10 MIN.)

Evaluate numerical expression with partner

1.4.7 WRAP-UP (5 MIN.)

Reflect on validity of statements related to lesson learning targets

1.4.2 REFRESHER (5-10 MIN.)

Number operations in correct order

1.4.4 COLLABORATION (10-15 MIN.)

Error analysis of samples of student work

1.4.6 YOUR TURN! (10 MIN.)

Evaluate numerical expressions independently

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Order of Operations

MATERIALS

- (Optional) Calculators
- (Optional) Cards for Card Sort
- (Optional) Scissors
- (Optional) Blank Paper
- (Optional) Glue

ADDITIONAL PREPARATION

- (Optional) 1.4.1 Warm-Up could be presented as a virtual tour.
- If structuring 1.4.3 Guided Practice as a card sort, print the tables in questions 1a and 1b and provide students with scissors, glue, and blank paper.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand that they should work from left to right when performing multiplication or division.
- Students may not understand that they should work from left to right when performing addition or subtraction.
- Students may not understand that the opposite of the absolute value of a number will be a negative number.

THINGS TO CONSIDER

- If students use PEMDAS to remember the order of operations, remind them that they should work from left to right with multiplication/division and addition/subtraction.
- Remind students to work from the inside out when evaluating expressions with absolute values or grouping symbols.
- Consider displaying an Anchor Chart with visuals and examples of the order of operations for students to reference.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 1.4.1 Warm-Up, students are shown a photograph of Castillo de San Marco. Ask students “What do you notice? What do you wonder?” Give students a few minutes to write down things they notice and things they wonder. After students have had a chance to write down their responses, ask several students to share things they noticed and wondered. Record these for all to see. Steer the conversation to focus on math-related topics.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 1.4.4 Collaboration, students have the opportunity to demonstrate their mastery of the learning target by identifying and correcting errors in samples of student work. Discussing and reflecting on samples of student errors will decrease the likelihood of students making similar errors themselves in the future.

DIFFERENTIATE INSTRUCTION

Consider structuring 1.4.3 Guided Practice as a card sort. Having students cut out and glue the cards will provide kinesthetic and visual entry-points. During 1.4.5 Activity, students initially work individually to evaluate four different expressions. Then, they calculate the sum of their four answers, which should be the same. Students share how they solved their problems and help each other identify any possible errors, providing an auditory entry-point.

Step	Expression	Explanation
4	0.45	<i>Evaluate the multiplication</i>
2	$-6.75 \div 15 \bullet (-1)$	Evaluate the exponent
1	$-6.75 \div 15 \bullet (-1)^5$	Original expression
3	$-0.45 \bullet (-1)$	<i>Evaluate the division</i>

Additional differentiation opportunities are denoted throughout the lesson guide.

1.4.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Notice and Wonder

After showing students a photograph of Castillo de San Marco, ask “What do you notice?” and “What do you wonder?” Give students a few minutes to write down things they notice and things they wonder. After students have had a chance to write down their responses, ask several students to share things they noticed and wondered. Record these for all to see. Steer the conversation to focus on math-related topics.

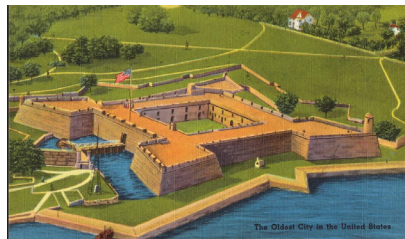
QUESTIONING

Prompt students to virtually tour the Castillo de San Marcos. The virtual tour will provide students with additional mathematical topics to notice and wonder about.



1.4.1 Warm-Up: Notice and Wonder

In 1672, the Castillo de San Marcos was built by the Spanish in St. Augustine, Florida to defend Florida and the Atlantic trade route.



1. What do you notice about the design of the Castillo de San Marcos?

Answers will vary. The design of the Castillo de San Marcos has a star shape. The middle piece is a square with 4 triangles at the points.

2. What do you wonder?

Answers will vary. How many entry points are there in the Castillo de San Marcos? How long did it take to build Castillo de San Marcos?

**1.4.2 Refresher: Order of Operations**

1. Work with a partner to number each operation so that the order of operations is in the correct order.

- 1 Perform operations inside grouping.
- 3 Perform multiplication or division in order from left to right. (→)
- 2 Evaluate expressions with exponents.
- 4 Perform addition or subtraction in order from left to right. (→)

1.4.2 REFRESHER: ORDER OF OPERATIONS**DIFFERENTIATE INSTRUCTION**

Consider creating and displaying an Anchor Chart with visuals and examples of the order of operations for students to reference.

TEACHER MOVES

If students use PEMDAS to remember the order of operations, remind them that they should work from left to right with multiplication/division and addition/subtraction.

1.4.3 GUIDED PRACTICE: USING THE ORDER OF OPERATIONS TO EVALUATE NUMERIC EXPRESSIONS

DIFFERENTIATE INSTRUCTION

Consider structuring this component as a card sort. Having students cut out and glue the cards will provide kinesthetic and visual entry-points.

TEACHER MOVES

Think-Pair-Share

Prompt students to initially try the problems independently. Then, ask them to discuss their thinking with a partner. Finally, call on a variety of students to share during the whole-class debrief discussion.

DIFFERENTIATE INSTRUCTION

Prompt students to refer to the order of operations Anchor Chart created during 1.4.2 Refresher.

TEACHER MOVES

In question 2a, students may incorrectly evaluate $-|-8.7|$ as 8.7. Remind students to work from the inside out when evaluating expressions with absolute values or grouping symbols. This will help them understand that the opposite of the absolute value of a number will be a negative number.



1.4.3 Guided Practice: Using the Order of Operations to Evaluate Numeric Expressions

- For each table, an expression is evaluated, but the steps are mixed up. Order the expressions by writing the numbered step next to each expression shown. Explain the steps in the column on the right. The first table has been started.

- Complete the table.

Step	Expression	Explanation
4	0.45	<i>Evaluate the multiplication</i>
2	$-6.75 \div 15 \cdot (-1)$	Evaluate the exponent
1	$-6.75 \div 15 \cdot (-1)^5$	Original expression
3	$-0.45 \cdot (-1)$	<i>Evaluate the division</i>

- Complete the table.

Step	Expression	Explanation
3	$ 5.76 - 17.75$	<i>Evaluate the exponent</i>
1	$ (9.2 - 11.6)^2 - 17.75$	Original expression
5	-11.99	<i>Evaluate the subtraction</i>
4	$5.76 - 17.75$	<i>Evaluate the absolute value</i>
2	$ (-2.4)^2 - 17.75$	<i>Evaluate the parentheses</i>

- Evaluate each expression using the order of operations.

- $$\begin{aligned} &(-5 + 4.5)^3 - |-8.7| \\ &(-0.5)^3 - |-8.7| \\ &-0.125 - 8.7 \\ &-8.825 \end{aligned}$$

- $$\begin{aligned} &\left(\frac{4.1-2.5}{-2}\right)^2 \times (-9) \\ &(-0.8)^2 \times (-9) \\ &0.64 \times -9 \\ &-5.76 \end{aligned}$$

- $$\begin{aligned} &-7.1 + 2|1.46 - 9.2| \\ &-7.1 + 2|-7.74| \\ &-7.1 + 2(7.74) \\ &-7.1 + 15.48 \\ &8.38 \end{aligned}$$



1.4.4 Collaboration: Using Order of Operations

1. Amiyah and David simplified the same expression using an order of operations and arrived at two different solutions. Their work is shown.

Amiyah	David
$-10 - 7(2.5^2) - -5\frac{3}{4} $	$-10 - 7(2.5^2) - -5\frac{3}{4} $
$-10 - 7(2.5^2) - 5.75$	$-10 - 7(6.25) - -5.75 $
$-10 - 7(6.25) - 5.75$	$-10 - 43.75 - -5.75 $
$-17(6.25) - 5.75$	$-10 - 43.75 + 5.75$
$-106.25 - 5.75$	-48
-112	

- Amiyah and David both made common mistakes. Circle the mistakes in the work.
- Write a note to Amiyah and David explaining the mistake(s) they made.

Amiyah	David
<i>Answers will vary.</i> <i>Amiyah remember when using the order of operations, you must multiply before you add or subtract. You subtracted $-10 - 7$ and your next step should have been $-7(6.25)$.</i>	<i>Answers will vary.</i> <i>David the rule for absolute value represents how far a number is for zero, so when simplifying the absolute value you should not multiply the sign on the outside. You should have simplified the absolute value to -5.75 instead of $+5.75$.</i>

- Evaluate the expression correctly.

$$\begin{aligned}
 &-10 - 7(2.5^2) - |-5\frac{3}{4}| \\
 &-10 - 7(2.5^2) - 5.75 \\
 &-10 - 7(6.25) - 5.75 \\
 &-10 - 43.75 - 5.75 \\
 &-59.5
 \end{aligned}$$

1.4.4 COLLABORATION: USING ORDER OF OPERATIONS

DIFFERENTIATE INSTRUCTION

Prompt students to circle or underline the mistakes. Also, consider providing students with sentence stems or answer choices to help them compose their notes to the sample students.

1.4.5 ACTIVITY: EVALUATING EXPRESSIONS

DIFFERENTIATE INSTRUCTION

Allowing partners to discuss how they each solved their problems, and giving students the opportunity to help each other identify any possible errors, provides students with an auditory entry-point.

ENRICHMENT

As students work, prompt them with the following:

- What order of steps did you take?
- Did you treat the absolute value symbols as grouping symbols and work from the inside out?
- How is cubing a negative number different from squaring it?
- Did you perform multiplication and division before addition and subtraction?
- Did you work from left to right with multiplication/division and addition/subtraction?



1.4.5 Activity: Evaluating Expressions

- Decide with your partner who is Partner A and who is Partner B.
 - Evaluate the problems in your column.

Partner A	Partner B
$12 - -6.5 + 3^3 $ $12 - -6.5 + 27 $ $12 - 20.5 $ $12 - 20.5$ -8.5	$9 - 5^2 - 8.4 $ $9 - 25 - 8.4 $ $9 - 16.6 $ $9 - 16.6$ -7.6
$\left(\frac{3 - 7.5}{15}\right)^2 \times 8$ $(-0.3)^2 \times 8$ 0.09×8 0.72	$\left(\frac{4.5 - 7.5}{5}\right)^2 \times -\frac{1}{2}$ $(-0.6)^2 \times -\frac{1}{2}$ $0.36 \times -\frac{1}{2}$ -0.18
$(-32) \div (-2)^4 \times (-3.6 - 1.2)$ $-32 \div 16 \times -4.8$ -2×-4.8 9.6	$(-0.74 \times 5) + 48 \div (-2)^2$ $-3.7 + 48 \div 4$ $-3.7 + 12$ 8.3
$ (30.5 - 5^2) \div 2 $ $ (30.5 - 25) \div 2 $ $ 5.5 \div 2 $ $ 2.75 $ 2.75	$ (23.5 - 4^3) \div 10 $ $ (23.5 - 64) \div 10 $ $ -40.5 \div 10 $ $ -4.05 $ 4.05

- Calculate the sum of the answers in each column from Part A.

Sum of Partner A's Answers	Sum of Partner B's Answers
$-8.5 + 0.72 + 9.6 + 2.75$ 4.57	$-7.6 + (-0.18) + 8.3 + 4.05$ 4.57

- Both columns should have the same sum. If that is not the case, work with your partner to find and correct any errors in your work.



1.4.6 Your Turn!

1. Evaluate each expression. Show the steps in your work.

a. $-4.5 - |(3 \times -1.6)^2|$
 $-4.5 - |(-4.8)^2|$
 $-4.5 - |23.04|$
 $-4.5 - 23.04$
 -27.54

b. $\frac{(-9^0 \times 0.2)^4}{-2}$
 $\frac{(1 \times 0.2)^4}{-2}$
 $\frac{0.2^4}{-2}$
 $\frac{0.0016}{-2}$
 -0.0008

c. $(-18.6) \div 6 \cdot (-2.3 + 5.4)^2$
 $(-18.6) \div 6 \cdot (3.1)^2$
 $-18.6 \div 6 \cdot 9.61$
 $-3.1 \cdot 9.61$
 -29.791

d. $[(-3)^1 \times (3.7 - 5.8)]^3$
 $[-3 \times -2.1]^3$
 $[6.3]^3$
 250.047

1.4.6 YOUR TURN!

TEACHER MOVES

Collect and Display

As students evaluate the expressions, collect a variety of students' work, and display them for all to see. Then engage students in a discussion about whether they agree or disagree and why.



1.4.7 Wrap-Up: Cool Down

1. For each statement, write whether you agree with it, disagree with it, or are unsure.
Answers will vary.

Statement	Agree/Disagree/Unsure
The more digits a number has, the larger it is.	<i>Disagree</i>
When you add two positive decimals together, the value is always larger.	<i>Agree</i>
To add exponents, their bases must be the same.	<i>Agree</i>
The product of two fractions is always a larger number.	<i>Disagree</i>
The sum of a positive and a negative number is always negative.	<i>Disagree</i>

1.4.7 WRAP-UP: COOL DOWN

TEACHER MOVES

Consider taking a poll (physically or digitally) to gauge student understanding of this lesson.

DIFFERENTIATE INSTRUCTION

As an extension, have students explain why they agree, disagree, or are unsure. Prompt students to include an example in their explanations.

DIFFERENTIATE INSTRUCTION

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.